DevOps 2TIN

Recap

DE HOGESCHOOL MET HET NETWERK

Elfde-Liniestraat 24, 3500 Hasselt, www.pxl.be

Recap

- 3 ways

THE FIRST WAY

CREATE A DELIVERY SYSTEM

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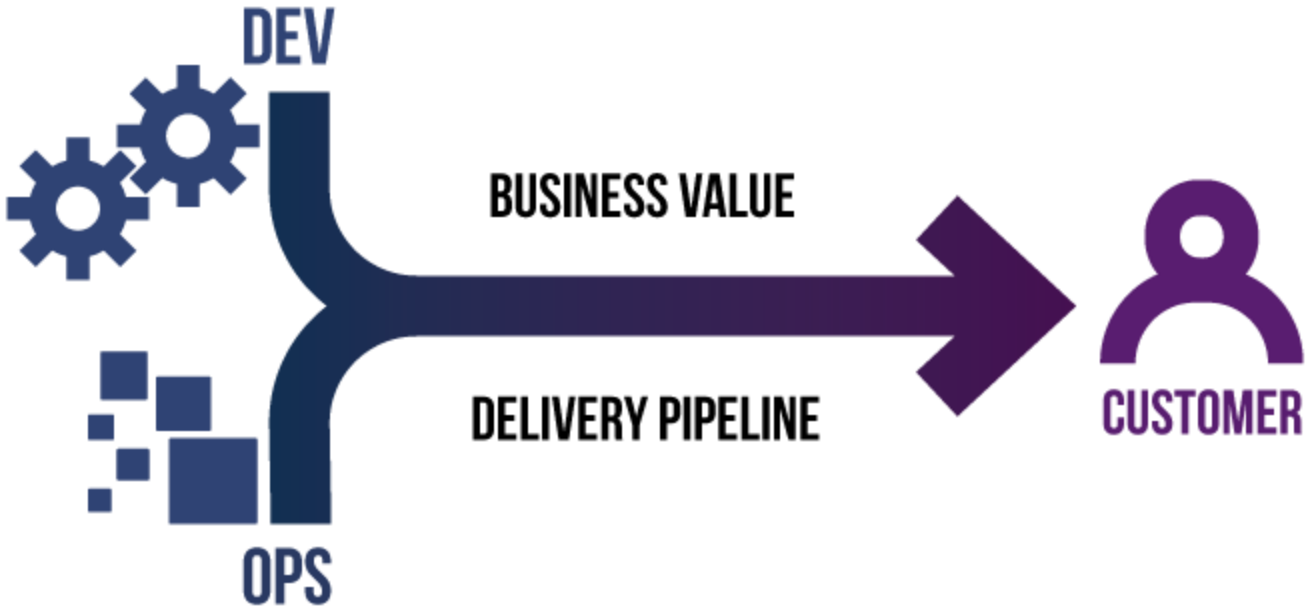
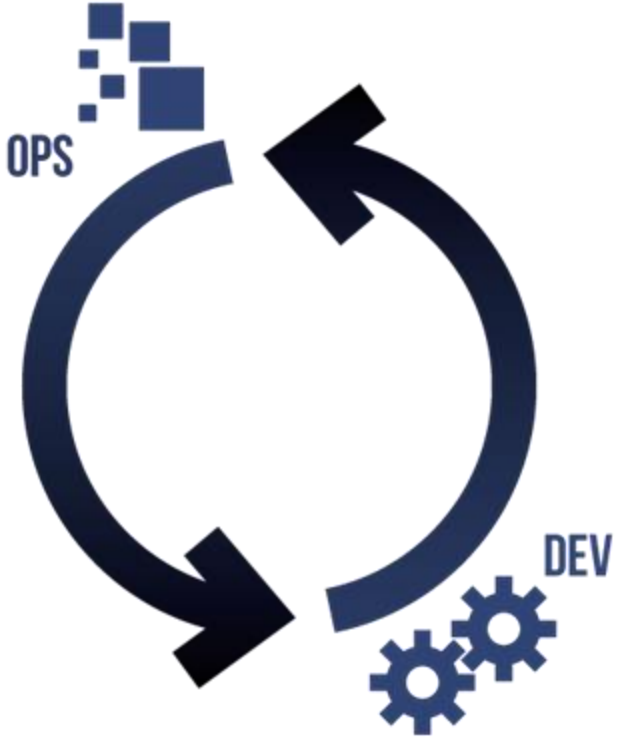


THE SECOND WAY

AMPLIFY FEEDBACK LOOPS

PXL DIGITAL



THE THIRD WAY

LEARN & EXPERIMENT CONTINUOUSLY

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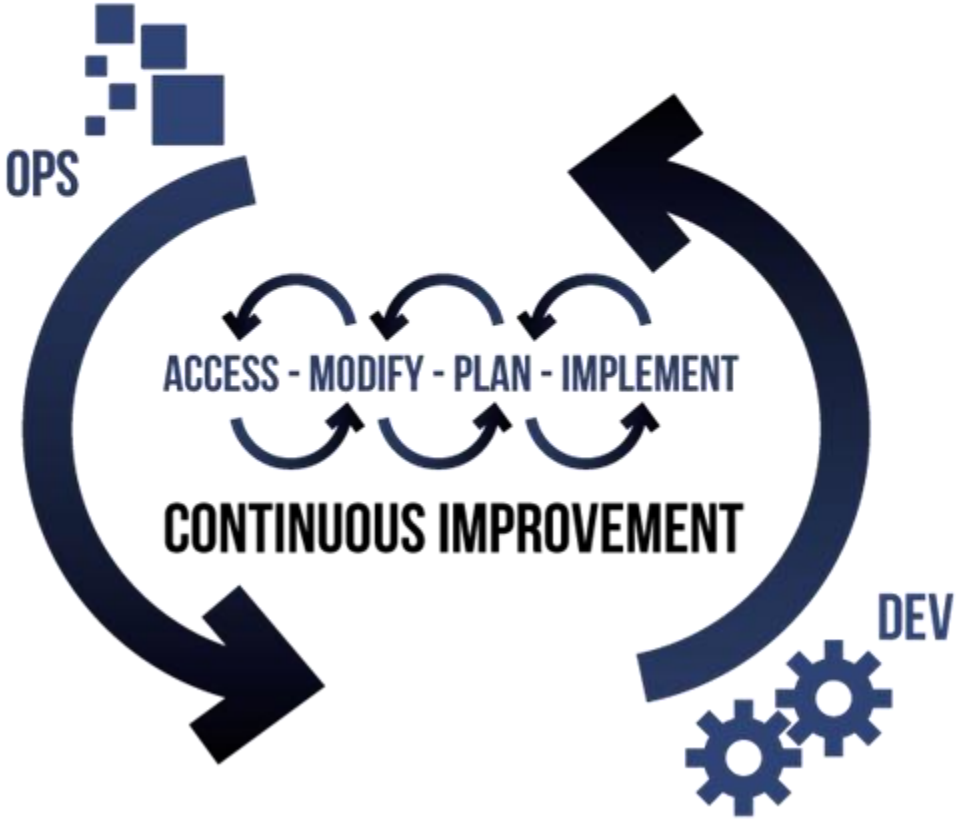

DevOps 2TIN

Chapter 3

Source control

DE HOGESCHOOL MET HET NETWERK

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Source control

Recap & general use
Git & Github
Branching
Github flow
Pull requests

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Waarom source control

- single source of truth (belangrijk binnen flow!)

Waarom source control

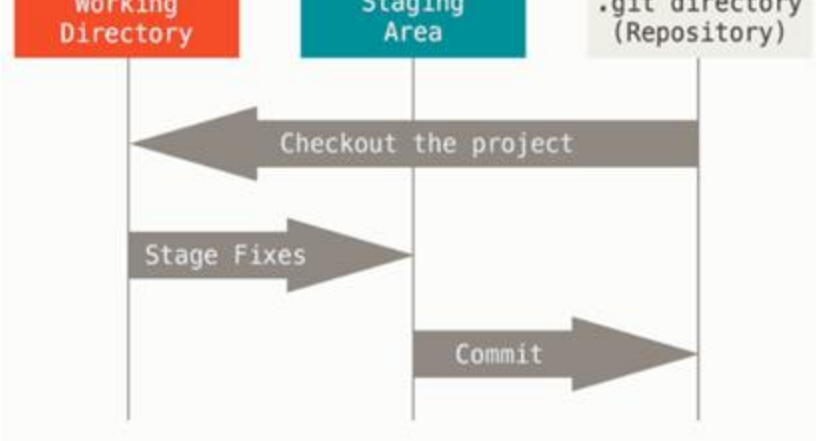
- Samenwerken als team aan dezelfde code

Waarom source control

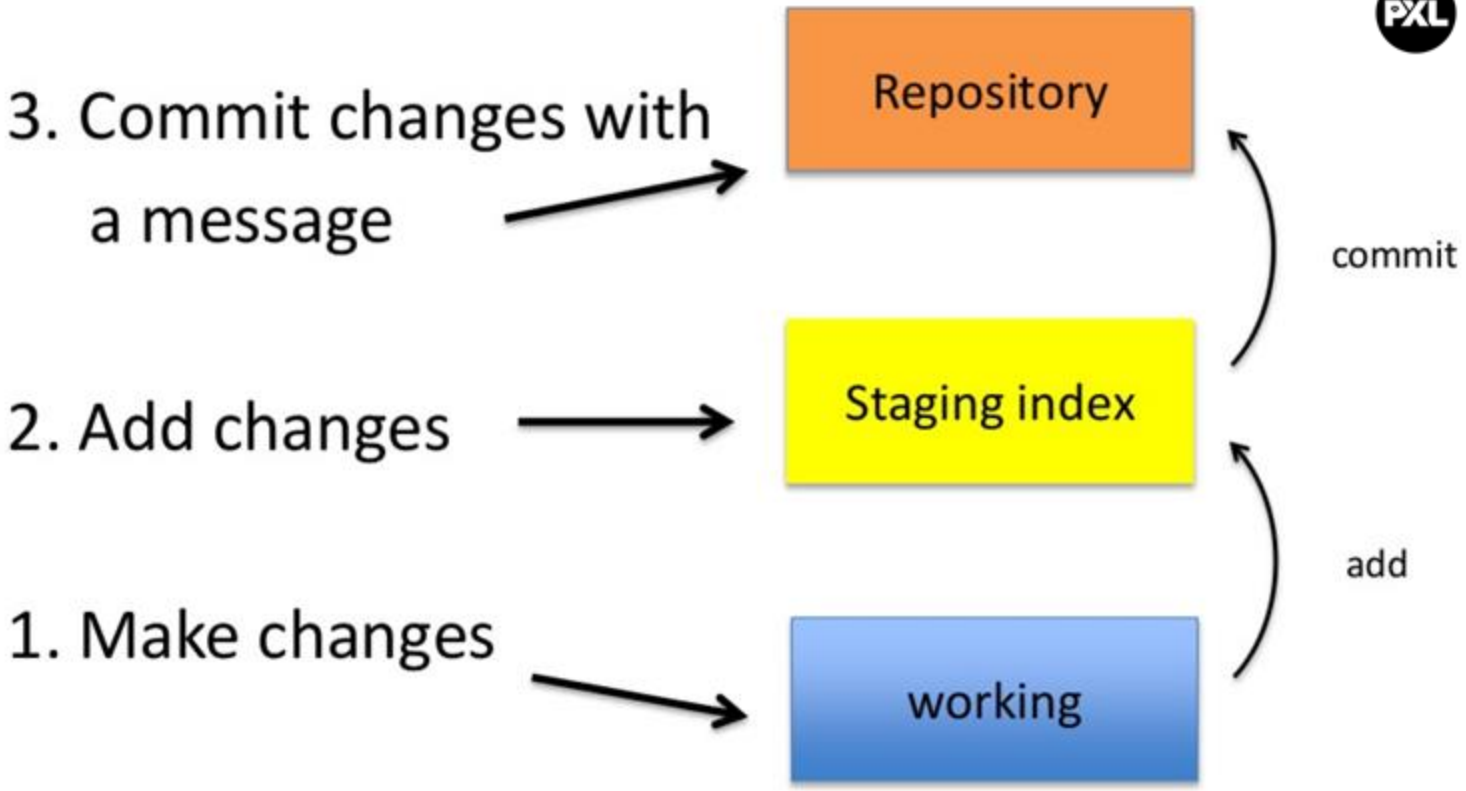
- Feedback over nieuwe releases van de code

Recap & general use

- Source control
- Waar voorzien:
  - Software code
  - Infrastructure as code
  - (tekst)documenten
  - ...
- Git, SVN, TFS, ...
  - ...



Recap & algemeen gebruik



Pluralsight - Git basics recap (G. Cleeren)

Recap & algemeen gebruik

- Commit messages
  - geeft **feedback** over wijzigingen
  - Duidelijke **flow** in aanpassingen code
  - https://chris.beams.io/posts/git-commit/
- Multi line commits
  - https://git-scm.com/book/en/v2/Customizing-Git-Git-Configuration

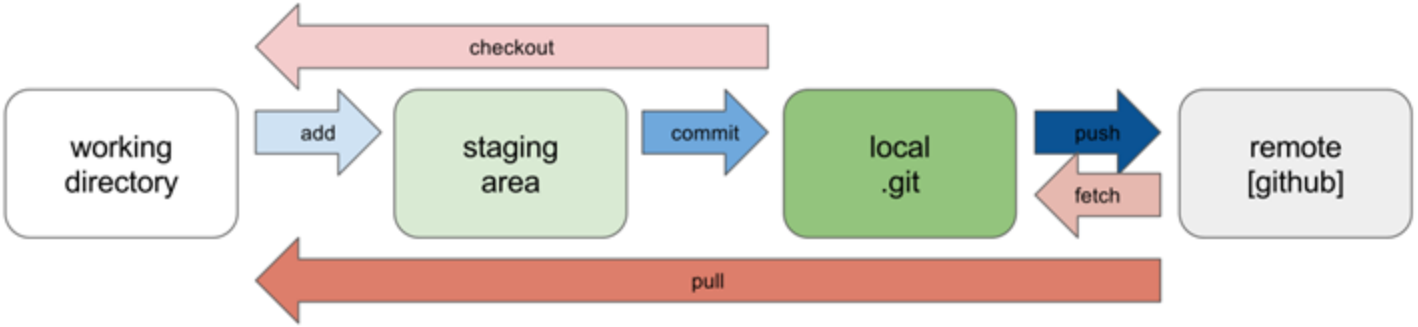
```
Subject line (try to keep under 50 characters)

Multi-line description of commit,
feel free to be detailed.

[Ticket: X]
```

Recap & algemeen gebruik

- Pushing code to a remote, central location



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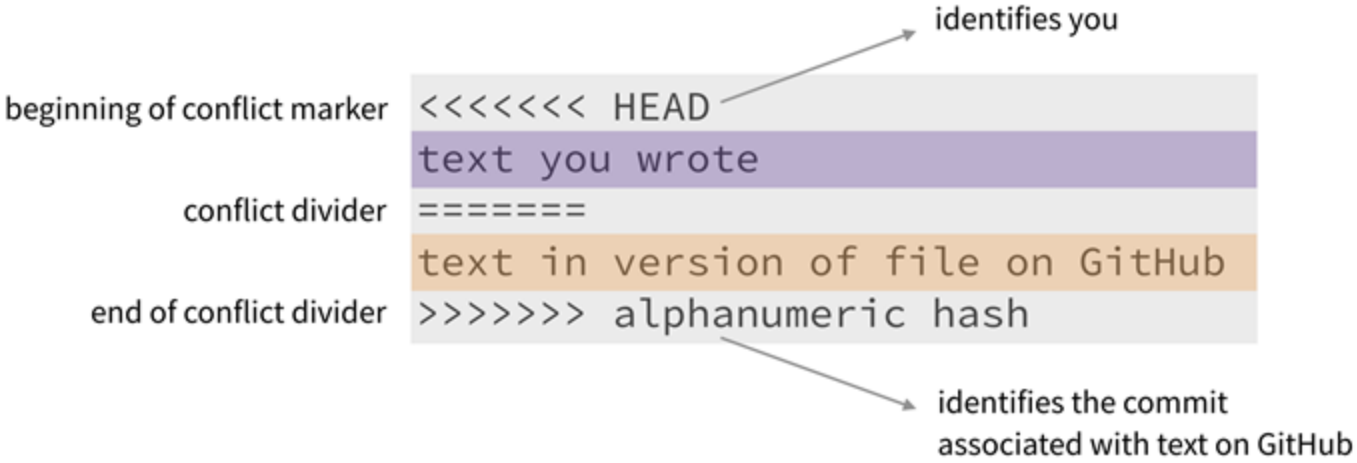
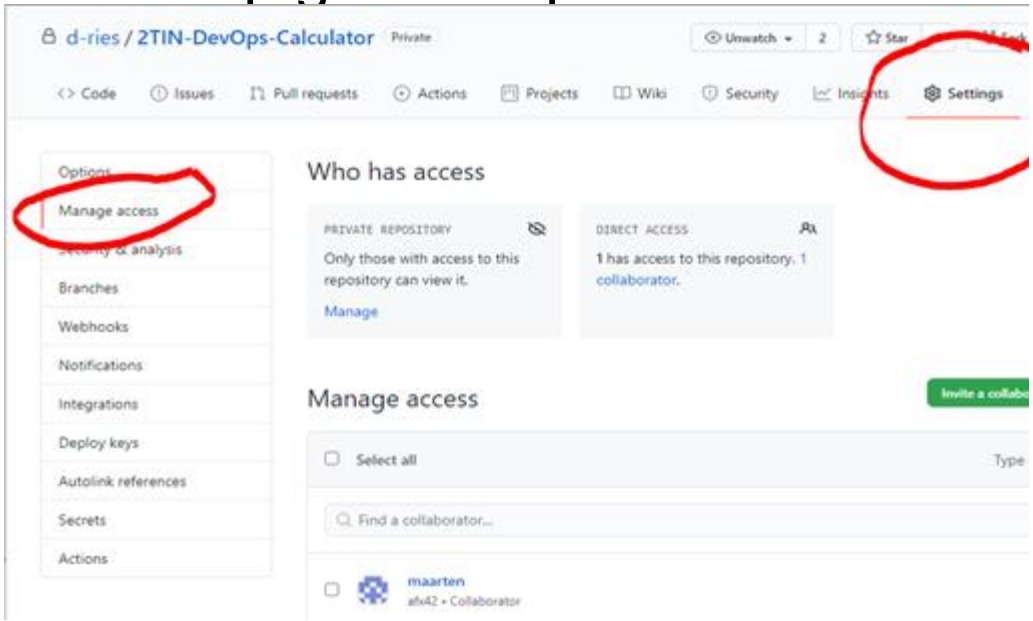
Recap & algemeen gebruik

- Merge conflicts
  - Changes on the same line
  - Changes in a deleted file



Github - Collaboration & permissions

- Samenwerken op github repositories



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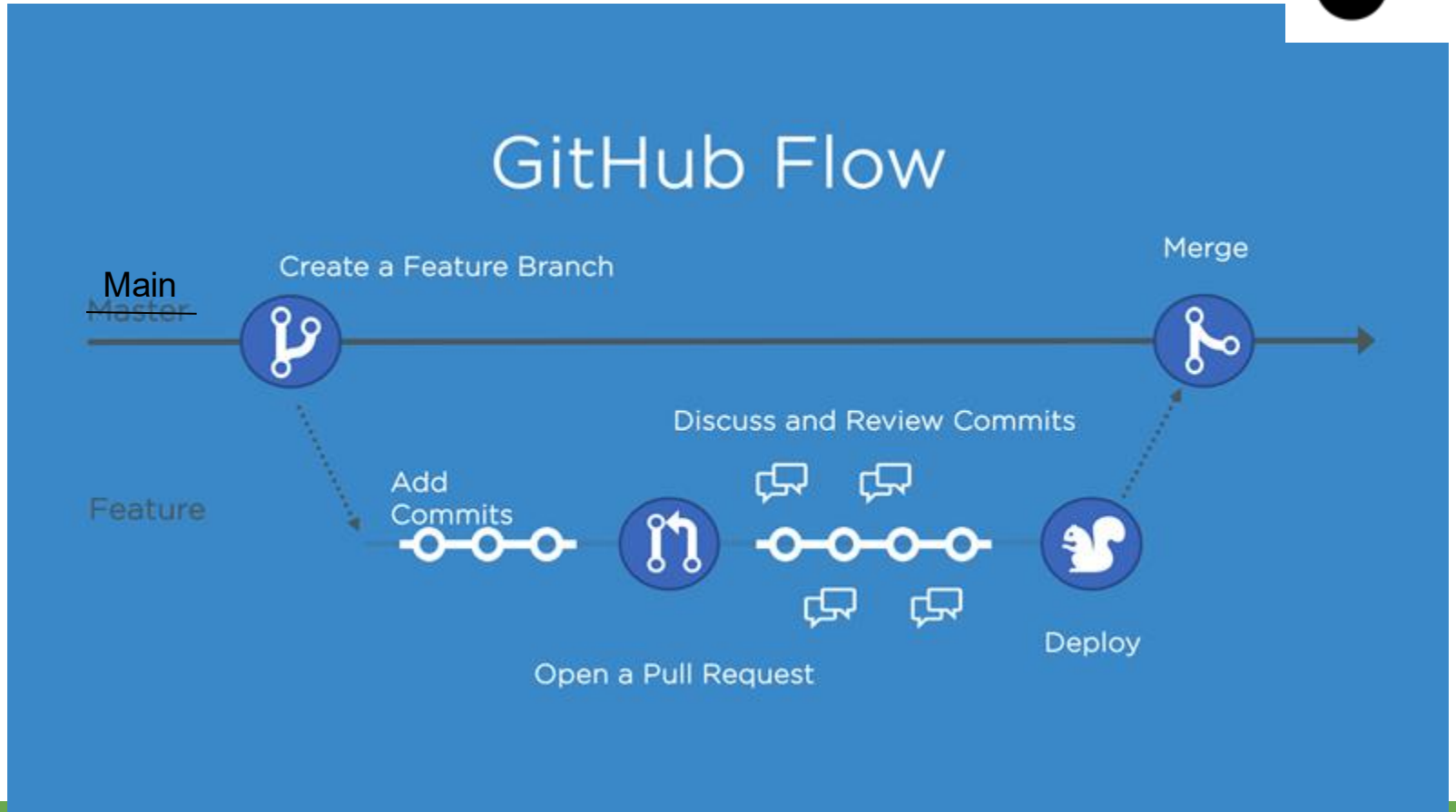
Branching strategie

- Strategie voor het (samen)werken met versiebeheer
  - Verschillende varianten
- Aftakkingen van de ~~master~~ main branch tijdens het ontwikkelen
- Duidelijke **Flow** doorheen het ontwikkelproces

https://guides.github.com/introduction/flow/ https://app.pluralsight.com/course-player?clipId=2b66962c-12b5-4e4c-9493-8d0c8f6d31ce

Github Flow


GitHub Flow
Main
~~Master~~
Create a Feature Branch
Merge
Discuss and Review Commits
Feature
Add
Commits
Open a Pull Request
Deploy

Github flow - Pull requests

1. Een developer maakt een feature in een aparte branch in zijn lokale repo
1. De developer pusht de branch naar de centrale repository op Github
1. De developer start een pull request via Github
1. De rest van het team evalueert de code, geeft feedback, past aan, …
1. De verantwoordelijke van de repository merged de wijzigingen naar de main branch en sluit de pull request

Github Flow - Branching

- Branches zijn aftakkingen
  - default branch = main / development branch

https://app.pluralsight.com/course-player?clipId=0c5b9d80-3bfd-4a73-ac56-96a17f532d25
https://www.atlassian.com/git/tutorials/using-branches

Github Flow - Branching

- Main branch = releases
- Development branch =

https://www.atlassian.com/git/tutorials/using-branches

Github Flow - Branching

- Branches maken

```
$ git branch nieuwe-branch
```

Of

```
$ git checkout -b nieuwe-branch
```

https://www.atlassian.com/git/tutorials/using-branches
https://app.pluralsight.com/course-player?clipId=75b80efe-3b43-4935-8d87-36da82ac778e

Github Flow - Branching

- Branches gebruiken
  - bestanden in working dir worden overschreven

```
$ git checkout branch-naam
```

- Branches bekijken

```
$ git branch -a
```

https://www.atlassian.com/git/tutorials/using-branches/git-checkout

Github Flow - Branching

- Branches pushen naar github

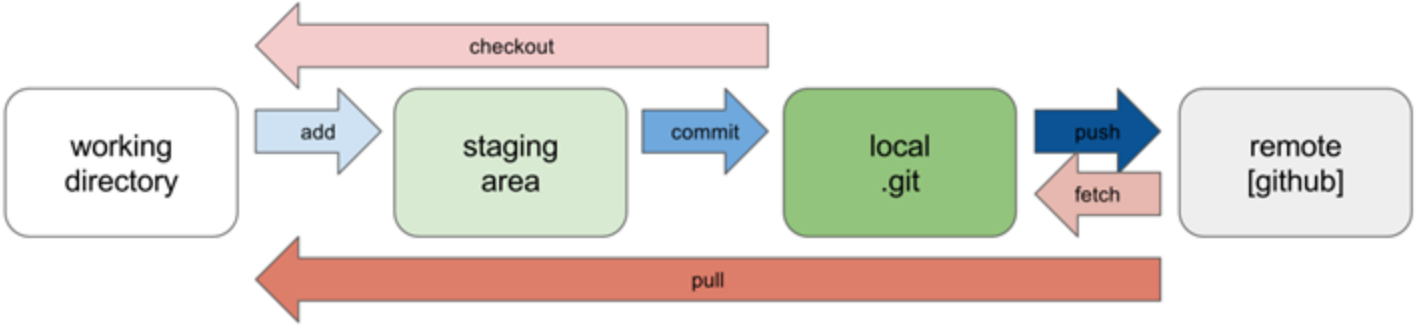
```
$ git push origin nieuwe-branch
```

- Branches pullen van github

```
$ git fetch --all
```

https://www.atlassian.com/git/tutorials/using-branches/git-checkout

Github Flow - Branching



https://www.atlassian.com/git/tutorials/using-branches/git-checkout

Github Flow - Branching

- Merging a branch
  - make sure you are on the main branch

```
$ git merge nieuwefeature
```

- Deleting branches

```
$ git branch -d nieuwefeature
```

Gebruik pull requests om een merge te doen!

https://www.atlassian.com/git/tutorials/using-branches/git-merge

Github flow - Pull requests

- Signaal geven dat code klaar is voor merge
- Waarom pull requests?
  - Evaluatie van code voor het mergen
  - Communicatie tussen developers, feedback voorzien
- Inhoud pull request:
  - comments, omschrijving, tag collega's, …

https://app.pluralsight.com/course-player?clipId=a486511a-b79e-48e9-8b4f-12b7e2ebd352
https://www.atlassian.com/git/tutorials/making-a-pull-request

Github flow - Pull requests

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d-ries / jenkins-test Private

Unwatch

Code | Issues | Pull requests | Actions | Projects | Wiki | Security

awsome-feature had recent pushes less than a minute ago

Compare & pull request

awsome-feature

Go to file | Add file | Code

This branch is 1 commit ahead of master.

Pull request | Compare

d-ries update dockerfile with whitespace ... now 21

src	test	6 months ago
Dockerfile	update dockerfile with whitespace	now
Jenkinsfile	test	6 months ago

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also compare across forks.

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base: master ← compare: awsome-feature ✓ Able to merge. These branches can be automatically merged.

update dockerfile with whitespace

Write | Preview

Description
Please include a summary of the change and which issue is fixed. Please also include relevant motivation and context. List any dependencies that are required for this change.

Fixes # (issue)
Type of change
Please delete options that are not relevant.

- [ ] Bug fix (non-breaking change which fixes an issue)
- [ ] New feature (non-breaking change which adds functionality)
- [ ] Breaking change (fix or feature that would cause existing functionality to not work as expected)

How Has This Been Tested?

Attach files by dragging & dropping, selecting or pasting them.

Create pull request

Reviewers
No reviews

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Linked issues
Use Closing keywords in the description to automatically close issues

Github flow - Pull requests

- Pull request templates
  - maak file pull_request_template.md aan in je root directory van de repository
  - Markdown syntax (md)

https://app.pluralsight.com/course-player?clipId=5bf0cb6a-3483-4f44-8b16-ee66a5a41acf

Github flow - Pull requests

- Commenting & giving feedback on pull requests

Code | Issues | Pull requests 1 | Actions | Projects | Wiki | Security | Insights | Settings

update dockerfile with whitespace #1

Edit | Open with

Open | d-ries wants to merge 1 commit into master from awsome-feature

Conversation 0 | Commits 1 | Checks 0 | Files changed 1 | +1 −1

d-ries commented 1 minute ago

Description

Deze pull request voorziet aanpassingen aan de dockerfile zodat er op het einde een whitespace staat.

@d-ries : Klopt het dat die daar moet staan?

Fixes # 123

Type of change

Please delete options that are not relevant.

- [ ] New feature (non-breaking change which adds functionality)

Reviewers
No reviews

Assignees
No one—assign yourself

Labels
enhancement

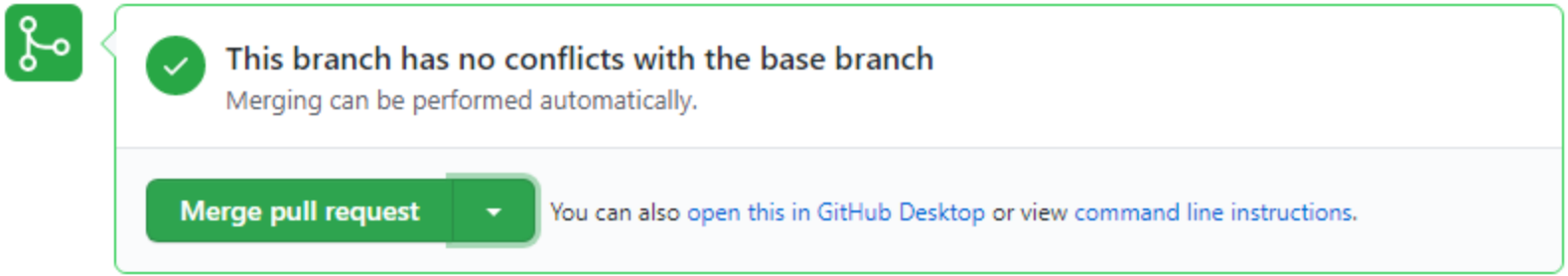
Projects
None yet

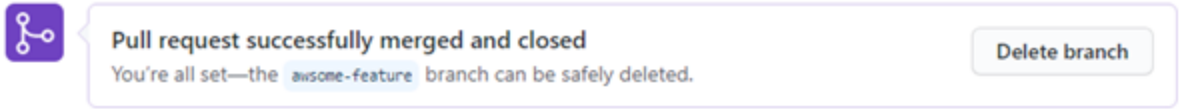
Milestone
No milestone

https://app.pluralsight.com/course-player?clipId=e23a835c-36a7-4e0f-9047-

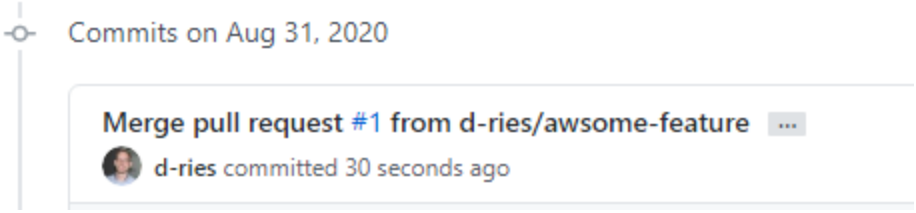
Github flow - Pull requests

- Merging pull requests
  - Approval van meerdere teamleden?
  - Merge conflicts?





Github flow - Pull requests



Assignments

Introductieopdracht
Source control

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